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TECHNO-ECONOMIC ANALYSIS OF OFF-GRID HYBRID PV-DIESEL-BATTERY SYSTEM IN KATSINA STATE, NIGERIA

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Abstract

Continuous burning of fossil fuel has a great impact on the environment. Thus, moving toward renewable energy system or hybridization of alternative energy systems can reduce the emission of hazardous gases produced by the fossil fuels. This paper presents a techno-economic analysis of hybrid PV/diesel and battery system configuration of five (5) Houses at Barhim Quarters, Katsina, Nigeria. HOMER optimization software was used for the techno-economic analysis. This paper shows how a meaningful amount of CO_2 can be reduce and the impact of PV penetration by the optimized hybrid PV/diesel system configuration. Having compared the configurations such as the PV-diesel-battery, PV-diesel without battery, Standalone diesel and standalone PV system based with $0.434/kW, $0.645/kW, $0.721/kW and $0.908/kW respectively as their cost of energy. Considering the Net present costs, cost of energy, CO_2 emission, excess of electricity generated and renewable penetration, a PV-diesel-battery Hybrid Power System (HPS) was proposed for the area under focus.

Keywords: Renewable energy, Cost of Energy, Emission, HOMER, Irradiance and Katsina

1.0 Introduction

The electrification of rural communities is one of the effective tool for sustainable development of such places in both developed and developing countries. Years back, a remarkable interest in the development of wind/PV, PV/diesel, wind/diesel as well as wind/PV/diesel ranging from medium scale to large scale hybrid power systems (HPSs) for the electrification of several rural communities across the world was observed (Rehman and Al-Hadhrami 2010). According to a survey conducted by the United Environment Program (UNEP), an estimated 1.7-2.0 billion people across the globe cannot access electricity service from the grid, and the majority of these people are dwellers of underdeveloped rural villages (Rehman *et al*. 2007, García-Valverde *et al.* 2009). The poor distribution of electrical energy is due to so many factors, to mention among are; the isolation of the rural villages from the grid where by extending grid to such sparsely populated areas is not economical and the harsh terrain of the location (Gabler 1998). This is one of the major reason behind the ongoing research on Hybrid Power Systems (HPSs) and Renewable Energy Resources (RESs). Another important factors include depletion of fossil fuel, increasing energy demand as a result of rapid global population growth. Electricity generation from solar via Photovoltaic (PV) arrays has the advantage of being free from emission and sustainable resources. Presently, the greatest solar panels has the ability of converting about 22% of the spectral rays from the sun into electricity, yet, huge amount of money and research effort are being channeled into making solar module technology as efficient as possible (Shaahid and Elhadidy 2008, Kumar and Manoharan 2014).

The integration of PV system with diesel generator offer several advantages and help in avoiding intermittent supply. PV technology has been identified as a technology that is friendly to the environment and requires no other energy than the sun light. The hybrid power system proposed in this study comprises of diesel generator, photovoltaic panels, battery bank for the purpose of storage and a DC-AC power converter. The Hybrid Power Systems (HPSs) is not limited to only PV/diesel/battery combination, other combination reported from several literatures include (Getachew 2010, Pragya *et al.* 2010, Salwan and Dihrab, 2010, Subodh *et al.* 2011, Chong 2011, Ngan and Tan 2012, Rehman *et al.* 2012, Ayong *et al.* 2013, Chong *et al.* 2013, Ghasemi *et al.* 2013, Abdelhamid 2014, Adaramola *et al.* 2014, Tao 2014). The criteria for the selection of the hybrid system for a particular location depends on the topography of the area under focus, load demand, availability of seasonal energy resources and energy storage cost etc. (Kaldellis *et al.* 2010, Khelif *et al.* 2012).

2.0 Solar energy Background

Nigeria is located in west Africa, bordered by Niger to the North, Benin to the west , Cameroon to the east, chad to the northeast and Atlantic ocean to the south. It lies between latitude 44^{o} 16\` and $13^{o}$ 53\` to the north of the equator and longitudinal 2^{o} 40\` and $14^{o}$ 24\` to the east of the Greenwich meridian. It has 932,768 square kilometer land area and a population of over 170 million (National Commission for Mass Literacy). Nigeria lies within a high sunshine belt and has a huge solar energy potentials. The mean annual average of total solar radiation ranges from 3.5kWh/m^2/day in the coastal latitudes to about 7Kwh/m^2/day along the semi-arid areas in the far North. On average, the country receives solar radiation at the level of about 19.8MJ/m^2/day. The daily sunshine hours has an annual average of 4 to 9 hours per day, increasing from Southern part to Northern part. If the solar modules were to be used to cover 1% of Nigerian land area, it is possible to generate more than one hundred time the current generation of the country (Sambo 2012).

Katsina is one of the 36 states of Nigeria, which was situated in the Northwest. It has 34 Local Government Areas and a population of over 5 million. God has endowed the state with renewable energy sources such as wind, solar, hydro and biomass. Due to its high solar irradiant, various companies are coming to Katsina to partnership with the State Government or investors - so as to generate electricity using solar PV. During the tenure of the past administration of Governor Ibrahim Shehu Shema, 2015, a German developers set to build a 30MW PV power plant in Katsina (Photon.info).

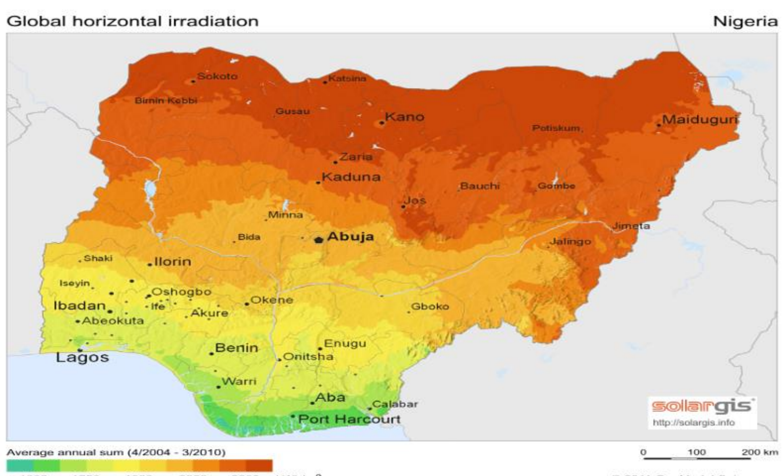


Figure 1: Global horizontal irradiation (GHI) of Nigeria(Aliyu *et al.* 2017)

From Figure 1, it is clear that, Katsina state has an average annual total solar radiation greater than 2200kWh/m^2

2.1 HOMER Simulation Software

HOMER software developed by the National Renewable Energy Laboratory (NREL) in United State was used in this study. The software was chosen as a result of its unrestricted availability for the modelling of micro grid and it simplicity of application. HOMER simulation tool significantly simplifies the task of scheming of hybrid systems whether stand-alone or grid connected. The algorithm of HOMER optimization and sensitivity analysis evaluate the techno-economic viability of large number of technology options as well as the account for the variation of technology cost, availability of energy resources and electrical load.

The hybrid optimization model for electric renewables (HOMER) software provide a detailed sequential simulation and optimization model. It is applicable to widespread scope of projects such as small and large scale power system. HOMER has the ability to model both the technical and economic aspects that were encompassed in this study, and finally, the optimized arrangement is subjected to a sensitivity analysis procedure in which numerous optimization each through a different set of input hypothesis is accomplished. HOMER model used to design and analyse the most efficient hybrid renewable energy systems for the proposed area. In the first phase of the study, solar resource of the studied area was evaluated. Then by using monthly solar profile, load demand values, economic constraints such as interest rate and project lifetime, the component characteristics such as inverter and battery sizes and cost and other related information were import to HOMER software. After the optimization analysis of the possible optimized hybrid energy configurations, the most economical and efficient hybrid energy combinations suggested by HOMER based on their total net present cost of the system and electrical production was proposed.

2.2 Net Present Cost (NPC)

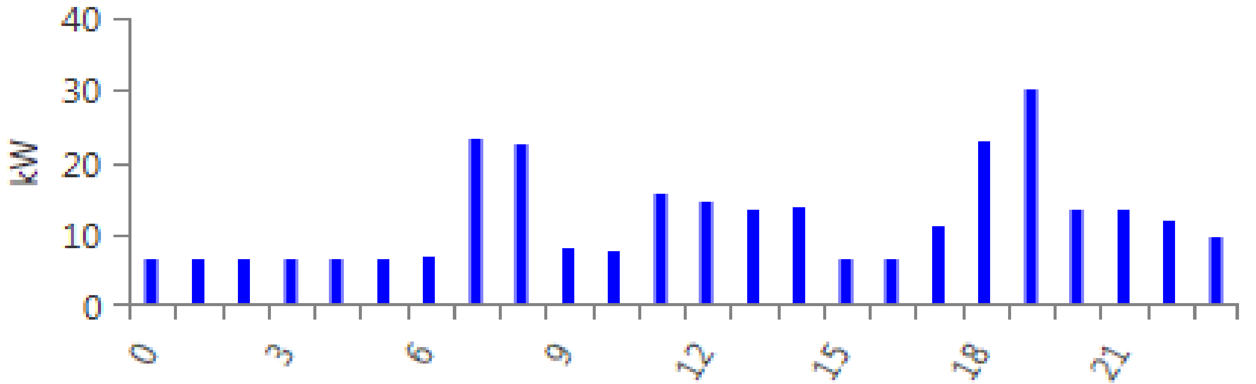
The quantity used to represent the life-cycle cost of the system is the total Net Present Cost (NPC). This single value includes all costs and revenues that occur within the project lifetime, with future cash flows discounted to the present. The total net present cost includes the initial capital cost of the system components, the cost of any component replacements that occur within the project lifetime, the cost of maintenance and fuel. The net present value of the system calculated using the formula

$$NPC = -C_0 + (B-C)\ {}^{T}{}_{E=1}\Sigma \left([1 + |^{d}/_{100}|]^{-6} + L_T\,[1 + |^{d}/_{100}|]^{-6}\right) \qquad 2.1$$

Where Co = initial investment, B= annual benefits, C= annual investment, d=discount rate, t=time. From among the mutually exclusive events, the project, which has maximum positive NPC, is economically more feasible.

3.0 Case study location

The paper presents a techno-economic analysis of PV/diesel hybrid system to supply a 5 Houses at Barhim Quarters Katsina, Nigeria. The case study is inside the city of Katsina - as the settlement is an urban area. Each house is a two bedroom flat, which consist of living room, master room, bedroom, three toilet, kitchen and store. The study area was chosen in order to encourage the use of distributed generation, alternative energy systems and to encourage the use of renewable energy. Figure 2 shows the daily load profile of 5 houses of Barhim Quarters Katsina, Nigeria.



Daily Load profile

Figure 2: Load profile of 5 houses of Barhim Quarters Katsina, Nigeria (HOMER 2017)

From the load profile in Figure 2, it can be observed that the peak load hour was between 19:00 – 20:00 hour due to many activities taken place such as cooking, pressing cloths, lighting, watching TV etc., the base load was between 23:00 – 06:00 hours due to less used of appliances as it is usually the time to rest or sleep. After inserting the load (kW) at each hour in a day, HOMER simulator software generate a scaled annual average demand of the case study as 288.93kWh/day which is the same as 105.4595MWh/year. The load type, wattage, usage hours and energy consumption of one house is depicted in Table 3.1a and 3.1b.

Table 3.1a: Power consumption of two Bedroom Flat House in Barhim Quarters Katsina, Nigeria

Appliances	Wattage	Usage (hours) Number	0	1	2	3	4	5	6	7	8	9	10	11
Freeze	800	1	800	800	800	800	800	800	800	800	800	800	800	800
Electric Cooker	1500	1	0	0	0	0	0	0	0	1500	1500	0	0	0
Lamp	40	29	320	320	320	320	320	320	400	200	100	0	0	0
Fans	80	4	160	160	160	160	160	160	160	160	160	160	160	160
TV	150	2	0	0	0	0	0	0	0	0	150	150	150	150
Laptop	80	2	0	0	0	0	0	0	0	0	80	80	0	0
Mobile charger	4	3	0	0	0	0	0	0	0	0	12	12	4	4
Blender	300	1	0	0	0	0	0	0	0	0	300	0	0	0
Radio	71	1	0	0	0	0	0	0	0	0	71	71	71	0
Pressing Iron	1000	1	0	0	0	0	0	0	0	0	1000	0	0	0
Satellite Dish	360	1	0	0	0	0	0	0	0	0	360	360	360	0
Electric Kettle	2000	1	0	0	0	0	0	0	0	2000	0	0	0	2000
Total power consumption per hour			1.28	1.28	1.28	1.28	1.28	1.28	1.36	4.66	4.533	1.633	1.545	3.114
Total power consumption for 5 houses			6.4	6.4	6.4	6.4	6.4	6.4	6.8	23.3	22.665	8.165	7.725	15.57

Table 3.1b: Power consumption of two Bedroom Flat House in Barhim Quarters Katsina, Nigeria

Appliances	Wattage	Usage (hours) Number	12	13	14	15	16	17	18	19	20	21	22	23
Freeze	800	1	800	800	800	800	800	800	800	800	800	800	800	800
Electric Cooker	1500	1	1500	1500	1500	0	0	0	1500	1500	0	0	0	0
Lamp	40	29	0	0	0	0	0	0	200	800	800	800	500	320
Fans	80	4	160	240	240	240	240	240	240	240	240	240	240	160
TV	150	2	150	150	150	150	150	150	150	150	300	300	300	150
Laptop	80	2	0	0	0	0	0	0	0	160	160	80	80	80
Mobile charger	4	3	4	4	8	8	8	8	8	8	12	12	8	8
Blender	300	1	300	0	0	0	0	0	300	0	0	0	0	0
Radio	71	1	0	0	71	71	71	0	0	0	0	71	71	0
Pressing Iron	1000	1	0	0	0	0	0	1000	1000	0	0	0	0	0
Satellite Dish	360	1	0	0	0	0	0	0	360	360	360	360	360	360
Electric Kettle	2000	1	0	0	0	0	0	0	0	2000	0	0	0	0
Total power consumption per hour			2.914	2.694	2.769	1.269	1.269	2.198	4.558	6.018	2.672	2.663	2.359	1.878
Total power consumption for 5 houses			14.57	13.47	13.85	6.345	6.345	10.99	22.79	30.09	13.36	13.32	11.795	9.39

3.1 Solar GHI resources

The case study has a latitude of 12058.91N and longitude of 70 37.31E. The solar GHI resource of the case study was downloaded online form NASA surface meteorology and solar energy in HOMER resource directory. The months, clearness index, daily radiation and a bar chart are shown from the downloaded data. The study has an annual average GHI of 5.94 kWh/m^2/day which is sufficient for solar PV installation. Figure 3 shows that from February to June the area has the high solar radiation because of the hot season period whereas December and August has the low solar radiation because of severe winter period and heavy rainfall respectively.

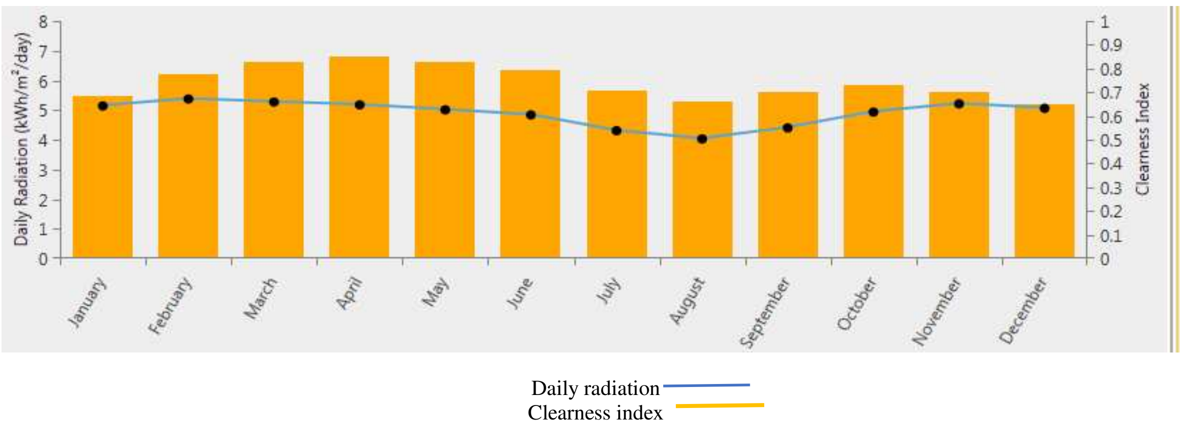


Figure 3: The annual solar radiation and the clearness index for the location of Katsina, Nigeria (HOMER, 2017).

3.2 Inverter Model

The inverter is selected in accordance with the overall capacity of the photovoltaic array. Therefore, an inverter with a rating of 50kW was considered in other to fully convert the maximum output of the photovoltaic module. However, the output from the photovoltaic module varies with sunlight intensity, hence the capacity of the converter was set to be 100 per cent and the overall efficiency was specified by the manufacturer as 97.5 per cent.

3.3 Battery Model

Proper sizing of battery is a key for the design of a reliable off-grid system. The main aim when sizing a battery bank is the installation of appropriate number of batteries to carry a certain load through a period during which the sun or wind is not accessible. The battery storage selected for the design was Trojan IND13-6V model. This battery was specifically designed to support renewable energy systems with big daily load and its high ampere-hour capacity is ideal for use in large off-grid photovoltaic (PV) system. Furthermore, it also met both IEC and BCI standard. The battery has a voltage rating of 2V with 1849A-H nominal capacity. Six batteries were connected in series to form a string which produces a 12V.

3.4 Diesel Generator

Another important module in off-grid system is the generator also referred to as gen-sets which are used to charge battery through period of low insolation. A diesel power generator set was in included in the design as an overall backup due to the fact that the duration of the sunlight tends to reduce during rainy season and furthermore, to supplementary energy system on days during which the solar irradiance is low. Cummins diesel powered generator was therefore selected for the purpose of this design. The selected generator has a power rating of 50kW. The generator was included to purposely operate occasionally whenever the designed storage battery bank cannot sufficiently supply the energy demand.

4.0 Results and Discussion

The simulation was conducted by comparing the optimal components configuration of a standalone diesel generator, PV/diesel with battery, PV/diesel without battery and PV battery. The diesel generator is varied for 30kW, 40kW, and 50kW. The PV capacity is varied for 30kW, 40kW, and 50kW. The energy storage device (battery) is varied for 50, 60 and 70 units and lastly the converter is varied for 45kW, 50kW, and 55kW.

4.1 Standalone PV system

Due to emission of hazardous gases by the diesel generator, this study will like to see the possibility of implementing standalone PV system. Even though PV modules and the battery are very expensive but it is 100% renewable energy penetration compared to the standalone diesel, PV-diesel without battery and PV-diesel with battery configurations. The schematic model of standalone PV system is shown in Figure 4.

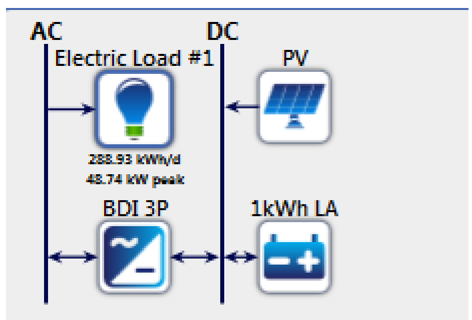


Figure 4: The schematic diagram of standalone PV system (HOMER, 2017).

After simulation of the standalone PV configuration, it was found out it has the highest excess electricity of 72.3% compare to the rest of standalone diesel, PV-diesel without battery and PV-diesel with battery configurations. It has the highest capital that constituent PV module $782,319, Battery $428,197 and power conditioning unit $26,455. One of its advantage is there is no emission of gases. The net present cost summary of standalone PV system is shown in Figure 5.

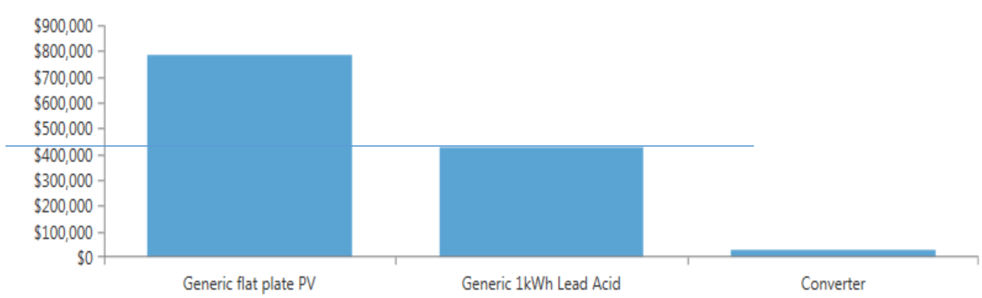


Figure 5: The Net Present Cost summary of Standalone PV system (HOMER, 2017)

The cost of electricity was computed to be $0.908/kWh which is higher than the standalone diesel system. Operating and maintenance cost is $24,906 which is less than the configuration of standalone diesel system. Another advantage of this configuration is no issue with fuel price. Due to oil crisis worldwide, the fuel price is unstable from time to time.

4.2 Standalone Diesel base system

The diesel generator capacity should be equal with the peak demand of the electric power. 50kW diesel generator was considered in this study in other to meet the peak demand of 48kW. The peak demand of 48kW was reached by considering the type of loads in all the five houses, wattage, usage hours and energy consumption of each house under consideration. The surplus of 2kW was considered as the spinning reserve. The operation cost of the generator was evaluated to be $74,140.10/year, levelized cost of energy (COE) 0.721 $/kW and net present cost (NPC) of $983,447 at a diesel price of 0.8 $/L. The generator has a life span of 15,000 operating hours. The schematic diagram of the configuration is shown in Figure 6.

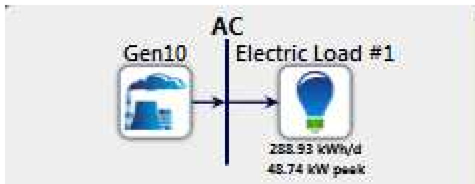


Figure 6: The systematic diagram of the standalone diesel system (HOMER, 2017)

4.3 Hybrid PV-diesel without battery system

The simulation results based on the system configuration have shown that 50kW PV array, 50kW diesel generator and 45kW converter are the optimized configuration i.e. the best configuration because of its lowest net present value NPV and cost of energy COE which are $879,550 and $0.645/kWh respectively. The systematic diagram of the hybrid PV/diesel without battery is shown in Figure 7.

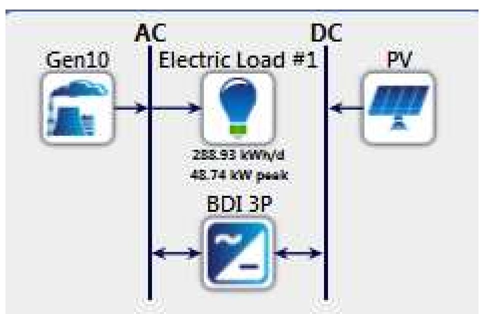


Figure 7: The systematic diagram of the hybrid PV/diesel without battery (HOMER, 2017)

Table 1 shows the production and fraction (%) of PV and diesel generator. The excess electricity is 40.9% which is higher than the configuration with battery and lower than PV/battery. The excess electricity from the PV is usually reduced due to the present of battery in the PV/battery system. Battery used a portion of that excess electricity for charging and supplement the load in some hours. Here the renewable energy penetration is 12.71%. However, the emission are higher than the PV/diesel with battery configuration as it is shown in Figure 7.

4.4 Hybrid PV-diesel-battery system

The sensitivity analysis result of the hybrid PV-diesel system with battery storage element is depicted in table 1, in this scenario, the influence of battery storage on the proposed hybrid power system was discussed. The complete system comprises of components such as PV array of 30kW capacity, 30kW diesel generator and 60 units of batteries. The schematic diagram of the configuration is shown in Figure 8.

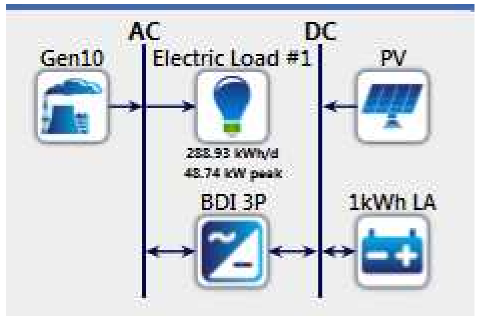


Figure 8: The systematic diagram of the hybrid PV/diesel with battery (HOMER, 2017)

Table 1 shows that the hybrid PV/diesel with battery storage has a total NPC of $286,315, this indicates that adding battery storage to the last previous configuration system has decrease the total net present cost even though, the addition of battery has increased the total capital and replacement cost, it also resulted to an annual reduction of $18,383/year in the cost of operation. Therefore, the total NPC of this system configuration has been reduced to $589,442 from $879,550 in the no-battery configuration discussed. Furthermore, the renewable penetration in this proposed system was found to be 31.8% that is 19.1% more than the PV/diesel without battery configuration.

Table 4.1: Sensitivity analysis result of PV-Diesel-Battery System

PV (KW)	Gen (KW)	Battery (Units)	Converter (KW)	COE ($/Kwh)	Total NPC ($)	Operating Cost ($/yr.)	Initial Cost ($)	Ren. Frac. (%)	Gen (Hours)	Diesel (L)
30	30	60	45	0.432	589,442	35,037	136,500	32	4,229	26,662
30	30	70	45	0.432	591,509	34,965	139,500	32	4,155	26,524
30	30	60	50	0.434	592,087	35,126	138,000	32	4,229	26,662
30	30	70	50	0.436	594,155	35,054	141,000	32	4,155	26,524
30	30	60	50	0.436	594,733	35,214	139,500	32	4,229	26,662

Since the capacity of PV array is less here compared with that of the PV/diesel without battery design, the PV array produces 53,284 kWh/year of electrical energy. In this configuration, some part of the surplus electricity is stored in the batteries which are used whenever the need arises, this has led to the increase in the fraction of renewable energy penetration for the proposed system. Also the generated electricity by the PV array contributed more to the supplying demand than the PV/diesel without battery, hence, the need for the generator is reduced and therefore the penetration of non-renewable source for electricity generation will be reduced.

It can be seen from table 1 that the batteries produced an electrical energy at 12,500 kWh/year, thereby reducing the surplus electricity from 40.9%to11.5% in the no-battery configuration system. The battery bank is charged by the source from the PV array, therefore, the PV array and the batteries concurrently produced 31.8% of the electricity generation from the system which is also the total renewable generation (fraction) from the entire system.

The diesel generator being one of the major component of the system, it is operated for 4,229 hours per year and consumes a diesel of 26,662 L/year to produce 71,933 kWh (68.2%) of electrical energy per year which is the remaining percent of the whole power generation.

Table 4.2: System components operation results and pollutant emissions for PV–diesel– battery system.

Quantity	Value	Units
Battery		
Energy in	15,624	kWh/year
Energy out	12,500	kWh/year
Losses	3,124.20	kWh/year
Annual throughput	13,975	kWh/year
Expected life	3.43	Year
Diesel generator		
Hours of operation	4,229	hrs./year
Number of start	1,142	starts/year
Operation life	3.55	Year
Marginal generation cost	0.23	$/kWh
Electrical production	71,933	kWh/year
Mean electrical output	17.01	kW
Max. electrical output	30.00	kW
Fuel consumption	26,662	L/year
Pollutant		
Carbon dioxide	70,211	Kg/year
Carbon monoxide	173.31	Kg/year
Unburned hydrocarbons	19.20	Kg/year
Particulate matter	13.07	Kg/year
Sulfur dioxide	141.00	Kg/year
Nitrogen oxides	1,546.40	Kg/year

Initially, it is noticed that the addition of battery to the configuration has significantly reduced the carbon dioxide emission in the proposed system which is as a result of the reduction in hours of operation of the diesel generator. The comparison of PV/diesel without battery, standalone PV system, PV/diesel with battery and standalone diesel based on the percentage of excess electricity, CO_2 emission and renewable energy penetration is shown in Figure 9.

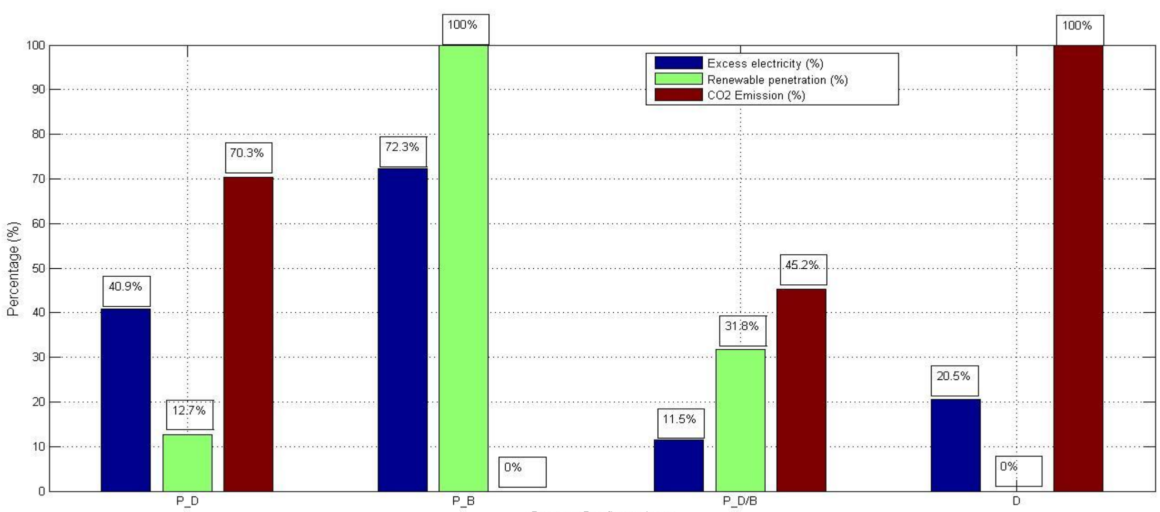


Figure 9: Comparison of four configuration system based on excess electricity, CO_2 emission and renewable energy penetration.

5.0 Conclusion

This study compared four different configuration namely; diesel generator which is the conventional system, PV-battery, PV-diesel and PV-diesel-battery configuration system. Hybrid optimization model for electric renewable (Homer) was used for the sensitivity analysis. The major focus on this study are the net present cost, renewable penetration and carbon dioxide emission as criteria's for determining the optimal system configuration. At the end of the study, a PV-diesel-battery hybrid power system (HPS) was proposed for the village under focus. The system comprises of 30 kW PV array, 30 kW diesel generator, 60 unit of battery and a 50kW power converter which produces 125,217 kWh/year in which 42.55% is renewable fraction. The system NPC is $589,442 for a life span of 25 years and 10% annual interest rate. Therefore, having consider the geography and the location of the case study which is inside the town of Katsina state. Hybrid PV/diesel with battery was suitable for that place so as to reduce the use of standalone diesel generator and to promote the use of alternative energy sources.

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